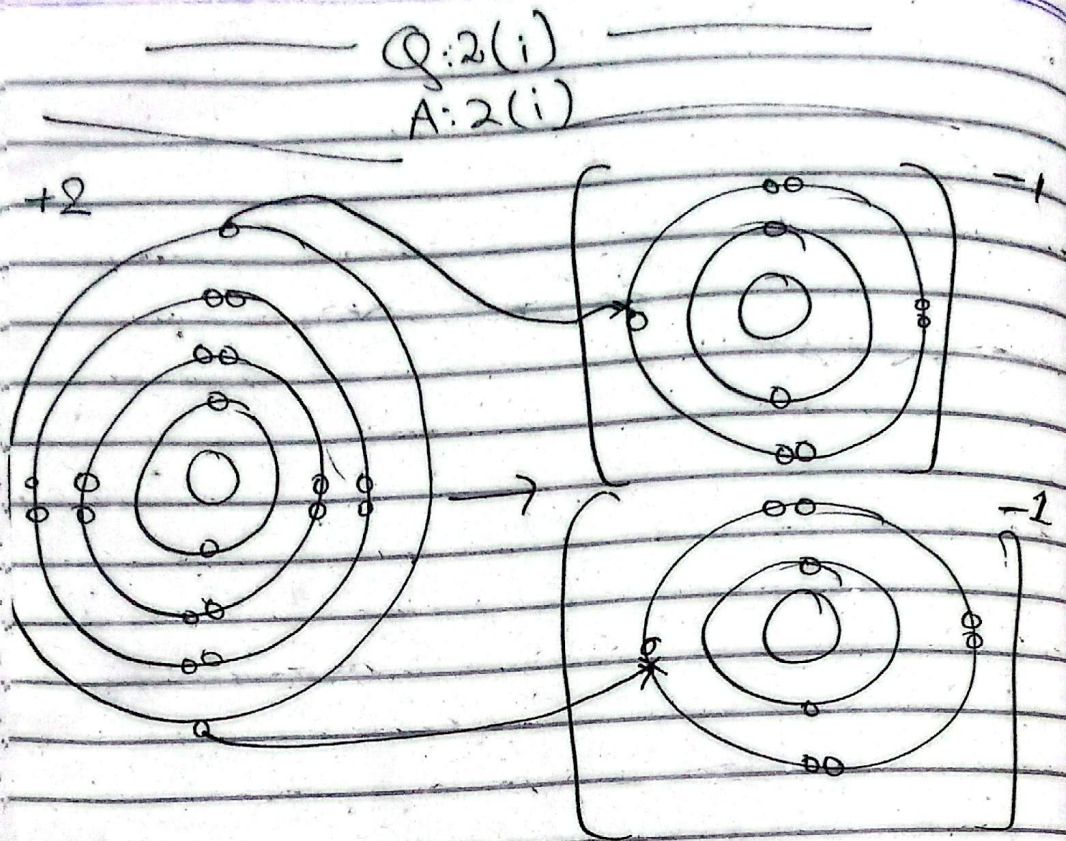


Chemistry

CAMPUS

MCQs :-

1. B
2. C
3. C
4. A
5. D
6. A
7. A
8. C
9. B
10. D
11. A
12. C
13. A
14. A
15. C
16. A
17. D
18. B
19. A
20. B
21. B
22. C
23. A
24. B



(b) Ionic bond : a bond formed by the electrostatic attraction b/w oppositely charged ions.

Q: 2(ii)
A: 2(ii)

(a) Relative atomic mass:-

$$= \frac{6 \times 7.5 + 7 \times 92.5}{100}$$

$$= \frac{692.5}{100}$$

$$= 6.925$$

(v)

Protons :- 3

electrons :- 3

neutrons :- 4

Q: 2(iii)

A: 2(iii)

(a) Electronic configuration of beryllium (2.4)

:-

$1s^2, 2s^2$

block :- s

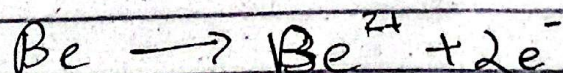
period :- 2nd

group :- 2

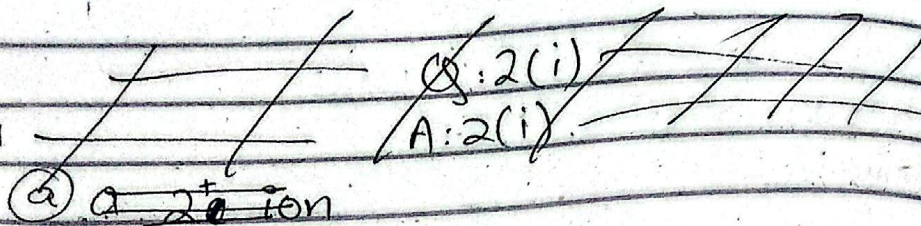
(i) Group (2) :- because beryllium has 2 electron in its shell.

(ii) Period :- (2) :- because it has 2 occupied electron shells.

(c) Be forms a Be^{2+} ion
It loses 2 electrons to achieve a stable electronic configuration :-



Since Be is in group 2, it has 2 electrons in its valence shell so it tends to lose both 2 electrons and forms 2^{+} ions.



~~Q: 3(i)~~
~~A: 3(i)~~
 (a) (i) Empirical formula :-
 The simplest whole number ratio of atoms of each element in a compound.

(ii) Molecular formula :-
 The actual number of atoms of each element in one molecule of a compound.

(b)
 K : 56.5
 C : 8.7
 O : 34.8

$$* K = \frac{56.5}{A.n} = \frac{56.5}{39} = 1.449$$

$$* C = \frac{8.7}{12} = 0.725$$

$$* O = \frac{34.8}{16} = 2.175$$

Now divide by smallest value:-

$$* K : \frac{1.449}{0.725} \approx 2$$

$$* C : \frac{0.725}{0.725} = 1$$

$$* O : \frac{2.175}{0.725} \approx 3$$

So ratio is

$$K : C : O$$

$$2 : 1 : 3$$

Empirical formula :- K_2CO_3

Molecular formula :-

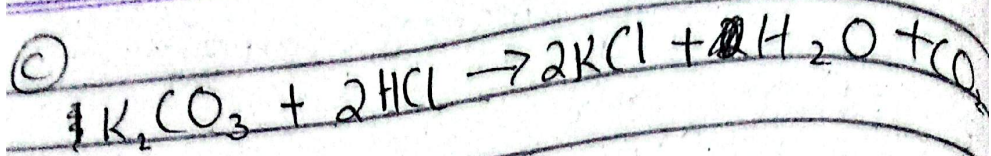
formula mass of K_2CO_3 :-

$$\begin{array}{r} 2(39) + 12 + 3(16) \\ 78 + 12 + 48 \\ 138 \end{array}$$

formula mass = 138

$$\text{multiplier} = \frac{138}{138} = 1$$

molecular formula :- K_2CO_3



Q: 3(ii)
a) molar mass of $Mg(NO_3)_2$

$$\begin{aligned} & \cancel{24} + \cancel{2(14)} + \cancel{6(16)} \\ & 24 + 2(14 + 3(16)) \\ & 24 + 2(14 + 48) \\ & 24 + 124 \\ & 148 \text{ g/mol} \end{aligned}$$

b) number of moles :- ?
molar mass = 148
mass :- 22.2 g

$$\text{moles} = \frac{\text{mass}}{\text{molar mass}}$$

$$\frac{22.2 \text{ g}}{148 \text{ g/mol}}$$

$$0.150 \text{ mol}$$

c) formula unit :- moles $\times$ Na

$$\begin{aligned} & 0.150 \times 6.022 \times 10^{23} \\ & = 0.9033 \times 10^{23} \\ & = 9.033 \times 10^{-1} \times 10^{23} \\ & = 9.033 \times 10^{22} \end{aligned}$$

1) Oxidation number of Mg.
+2

Rule:- Mg is a group 2 metal,
so it normally forms a +2
by losing its 2 outer electrons.

~~Q. Physics:-~~

MCQs

1. C

2. A

3. D

4. C

5. A

6. A

7. D

8. B

9. C

10. D

11. A

12. A

13. C

14. A

15. D

16. B

17. A

18. C

19. B

20. D

21. D

22. C

23. C

24. D